

Algorithmic Game Theory

Final Project

Exploring Voting Axioms & Impossibility

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 GitHub Repository

1 Introduction

The following paper explores the mechanisms and decision structures underpinning contemporary voting systems. Democracies strive to accurately represent the interests of their constituencies, with election results functioning as an aggregation of the individual voter preference profiles compromising the electorate. However, the various attributes (axioms) societies would like their electoral mechanisms to embody - anonymity, neutrality, transitivity, etc. - constrain it in ways that not every ideal attribute can be true simultaneously.

We begin by exploring several voting axioms that mechanism designers seek to achieve in an election system. We then walk through two experiments¹ that illustrate foundational impossibility results in social choice theory: Arrow's Impossibility Theorem and the Gibbard–Satterthwaite Theorem. The first experiment provides an exact enumeration of all social welfare functions satisfying Independence of Irrelevant Alternatives in a minimal setting, demonstrating how the additional requirements of unanimity and transitivity collapse the space of admissible rules to dictatorship. The second experiment adopts a complementary, empirical approach by stress-testing common voting rules for vulnerability to strategic manipulation, estimating how frequently a single voter can profitably misreport their preferences.

2 The Axioms

Axioms are well-defined properties of voting rules that allow for the comparison between different rules. They specify how a voting rule should behave across all possible preference profiles, such as fairness, consistency, or resistance to manipulation. In his paper *A Difficulty in the Concept of Social Welfare*, Kenneth Arrow lays out several axioms as constraints to collective decision making[Arr50]. Political and social scientist since have expanded upon and used axioms to study how collective decisions can (and cannot) aggregate individual preferences under realistic political conditions.

The following axioms form the core of contemporary voting theory[Özy21].

2.1 Pareto Principle (P)

A social welfare function F satisfies the *Pareto Principle* if for all preference profiles $(\succ_1, \dots, \succ_n) \in \mathcal{R}^n$ and all $x, y \in X$, whenever $x \succ_i y$ for all individuals $i \in \{1, \dots, n\}$, it follows that $x \succ y$ under the social ordering $F(\succ_1, \dots, \succ_n)$.

2.2 Anonymity (A)

A social welfare function F satisfies *Anonymity* if for every preference profile $(\succ_1, \dots, \succ_n)$ and every permutation π of the set of individuals,

$$F(\succ_1, \dots, \succ_n) = F(\succ_{\pi(1)}, \dots, \succ_{\pi(n)}).$$

That is, the social ranking is invariant to relabeling of individuals.

¹The replication code for both experiments is available in the corresponding GitHub repository. AI was used to build the code structure and finetune.

2.3 Neutrality (N)

A social welfare function F satisfies *Neutrality* if for every preference profile $(\succ_1, \dots, \succ_n)$ and every permutation σ of the set of alternatives X , the social ordering ranks alternatives in the same way after relabeling. Formally, for all $x, y \in X$,

$$x \succ y \text{ under } F(\succ_1, \dots, \succ_n) \quad \text{if and only if} \quad \sigma(x) \succ \sigma(y) \text{ under } F(\sigma(\succ_1), \dots, \sigma(\succ_n)).$$

2.4 Independence of Irrelevant Alternatives (IIA)

A social welfare function F satisfies *Independence of Irrelevant Alternatives* if for all $x, y \in X$ and all preference profiles $(\succ_1, \dots, \succ_n)$ and $(\succ'_1, \dots, \succ'_n)$, whenever for every individual i the relative ranking of x and y is the same under $\succ_i$ and $\succ'_i$, the social ranking of x and y is also the same under $F(\succ_1, \dots, \succ_n)$ and $F(\succ'_1, \dots, \succ'_n)$.

2.5 Ordinality (O)

A social welfare function F satisfies *Ordinality* if the social ordering depends only on the ordinal rankings of alternatives and not on any cardinal representation. That is, for any two preference profiles that induce the same ordinal rankings for all individuals, F produces the same social ordering.

2.6 Decisiveness (D)

A social welfare function F satisfies *Decisiveness* if for every pair of distinct alternatives $x, y \in X$ and every preference profile $(\succ_1, \dots, \succ_n)$, the social ordering produced by F ranks either $x \succ y$ or $y \succ x$. That is, the social preference relation is complete and admits no ties. Elections should result in a clear cut winner.

3 Arrow's Impossibility Theorem

Arrow's Impossibility Theorem states[Arr50; Mag25]:

When there are at least three alternatives, no voting rule can convert individual preference rankings into a collective social ranking while simultaneously satisfying a set of seemingly reasonable fairness axioms—unless the rule is dictatorial.

Essentially, when accounting for all voting axioms, fair aggregation of preferences is impossible. Arrow's theorem holds that dictatorship is not an arbitrary assumption, but the inevitable outcome of combining otherwise reasonable normative constraints (our axioms).

We present a complete enumeration of social welfare functions satisfying the IIA axiom for a minimal nontrivial setting with $n = 2$ voters and $m = 3$ alternatives. The purpose of this experiment is to make Arrow's Impossibility Theorem explicit by directly counting how many aggregation rules survive as we impose additional axioms.

A social welfare function maps every profile of individual preference rankings to a single social ranking of the alternatives. The IIA axiom requires that the social preference between any two alternatives a and b depend only on the individual preferences between a and b , and not on the presence or ranking of any third alternative.

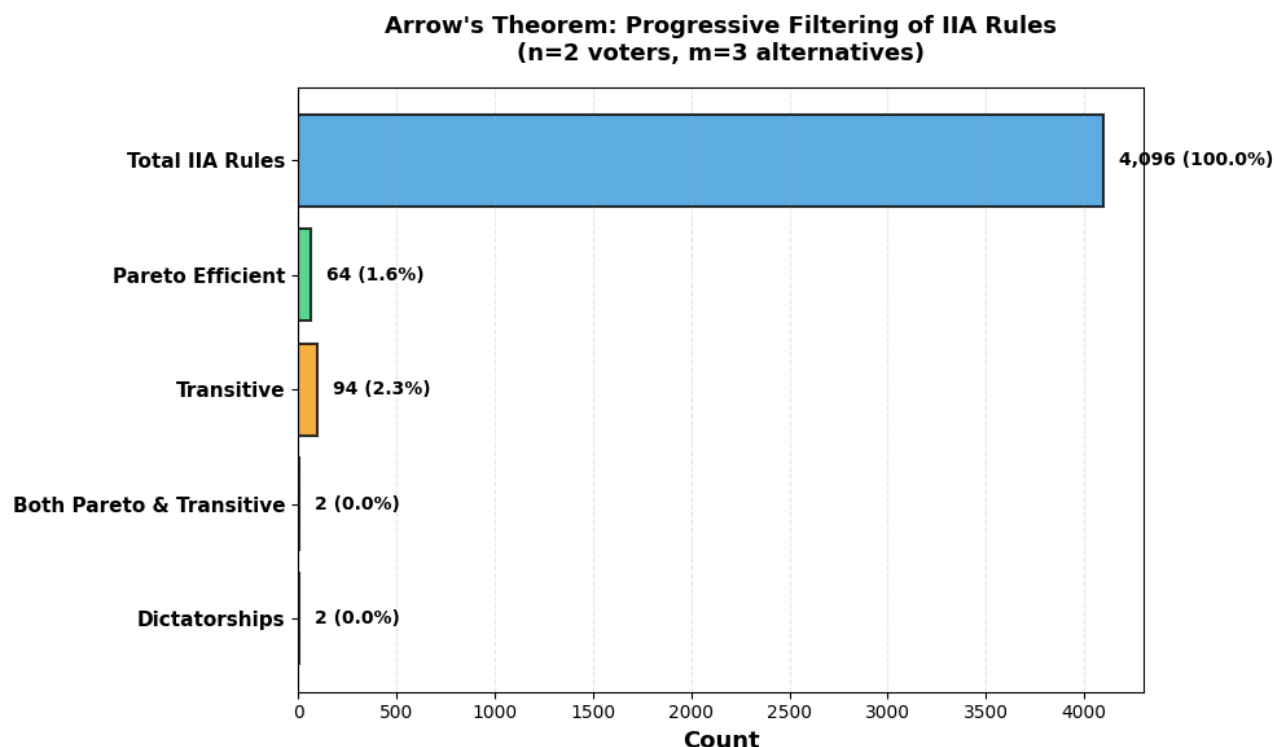


Figure 1: Imposing Pareto efficiency eliminates over 98% of all IIA-consistent aggregation rules.

Under IIA, the social preference between each unordered pair of alternatives can be specified independently. With two voters, each voter either prefers a to b or b to a , yielding $2^2 = 4$ possible preference profiles for each pair. Consequently, for any given pair (a, b) , an IIA-consistent social decision rule corresponds to a Boolean function $f_{ab} : \{0, 1\}^2 \rightarrow \{0, 1\}$, where the output indicates whether society prefers a to b . There are $2^4 = 16$ such functions.

With three alternatives, there are three unordered pairs of alternatives. Because IIA allows these pairwise rules to be chosen independently, the total number of IIA-consistent social welfare functions is $16^3 = 4,096$. These functions constitute the universe represented by the top bar in Figure 1.

We then impose additional axioms as filters on this universe. First, we test Pareto efficiency, which requires that if all voters rank alternative a above alternative b , then society must also rank a above b . Only 64 of the 4,096 IIA-consistent rules satisfy this unanimity condition, representing approximately 1.6% of the original universe.

Second, we test transitivity of the social ranking. A rule is transitive if it never produces a cyclic social ordering. Among the IIA-consistent rules, only 94 satisfy transitivity across all possible preference profiles, corresponding to approximately 2.3% of the universe.

When both Pareto efficiency and transitivity are imposed simultaneously, only 2 social welfare functions remain. These two rules are precisely the two dictatorships, one for each voter, in which the social ranking always coincides with the ranking of a fixed individual.

No non-dictatorial social welfare function satisfies Independence of Irrelevant Alternatives (IIA), Pareto efficiency (P), and transitivity simultaneously in this setting. The dramatic reduction from 4,096 admissible IIA rules to only 2 surviving rules provides a direct computational

demonstration of Arrow’s Impossibility Theorem. Dictatorship is not assumed in the construction but it emerges as the only possible outcome once the axioms are jointly imposed.

3.1 Gibbard–Satterthwaite Theorem

A compliment to Arrow is the Gibbard Satterthwaite theorem[Nar14; Mag25]:

Suppose there are at least three alternatives and that for each individual any strict ranking of these alternatives is permissible. Then the only unanimous, strategy-proof social choice function is a dictatorship.

Whereas Arrow’s theorem concerns the logical incompatibility of certain axioms for social welfare functions, the Gibbard–Satterthwaite Theorem characterizes the strategic vulnerability of non-dictatorial voting rules. Specifically, it states that for any deterministic, non-dictatorial social choice function with at least three alternatives, there exist preference profiles in which a voter can benefit by misreporting their true preferences.

The experiment evaluates the extent to which common voting rules are susceptible to profitable unilateral manipulation. We consider settings with $m = 4$ alternatives and either $n = 5$ or $n = 9$ voters. For each voting rule and each randomly generated preference profile, we select a single voter and examine whether that voter can submit an alternative ranking that yields a strictly preferred social outcome relative to truthful reporting, holding the preferences of all other voters fixed.

While less relevant for elections with large electorates, the Gibbard–Satterthwaite theorem highlights the inherent strategic vulnerabilities of non-dictatorial voting rules in small or medium-sized committees where individual incentives can be effectively influenced to affect voting outcomes.

A manipulation is deemed *profitable* if the outcome induced by the misreported ballot is strictly higher in the manipulating voter’s true preference ranking than the outcome under truthful reporting. For each rule and parameter setting, we estimate the share of profiles in which such a profitable unilateral deviation exists. All results are computed via Monte Carlo simulation over a large number of random preference profiles.

Figure 2 reports the estimated manipulation rates for six voting rules. Dictatorship is included as a benchmark and exhibits a manipulation rate of zero by construction, since the dictator can never benefit from misreporting and no other voter can influence the outcome. This confirms the theorem’s characterization of dictatorship as the only strategy-proof deterministic rule.

Among non-dictatorial rules, substantial manipulation rates are observed. Plurality and Borda exhibit the highest susceptibility, with profitable manipulations arising in approximately 40–50% of profiles. This reflects the fact that both rules are highly sensitive to strategic misrepresentation of rankings, particularly in competitive elections. Copeland displays a lower but still substantial manipulation rate of approximately 27–29%, indicating that pairwise majority-based methods mitigate but do not eliminate incentives for strategic voting.

Instant-runoff voting and two-round runoff perform better in this metric, with manipulation rates in the range of 18–24%. Nevertheless, these rates remain far from zero, illustrating that the use of full rankings or elimination procedures does not guarantee strategy-proofness. Importantly, increasing the number of voters from $n = 5$ to $n = 9$ does not eliminate strategic opportunities, suggesting that manipulation is not merely a small-electorate artifact.

Taken together, these results provide a quantitative illustration of the Gibbard–Satterthwaite Theorem. While the theorem is existential in nature, our experiment shows that strategic manipulation is not rare or pathological for commonly used voting rules. Instead, opportunities for profitable misrepresentation arise in a substantial fraction of preference profiles. When viewed alongside the results for Arrow’s theorem, this highlights a fundamental tension in social choice: rules that avoid dictatorship must either violate normative axioms such as independence of irrelevant alternatives or expose voters to incentives for strategic manipulation.

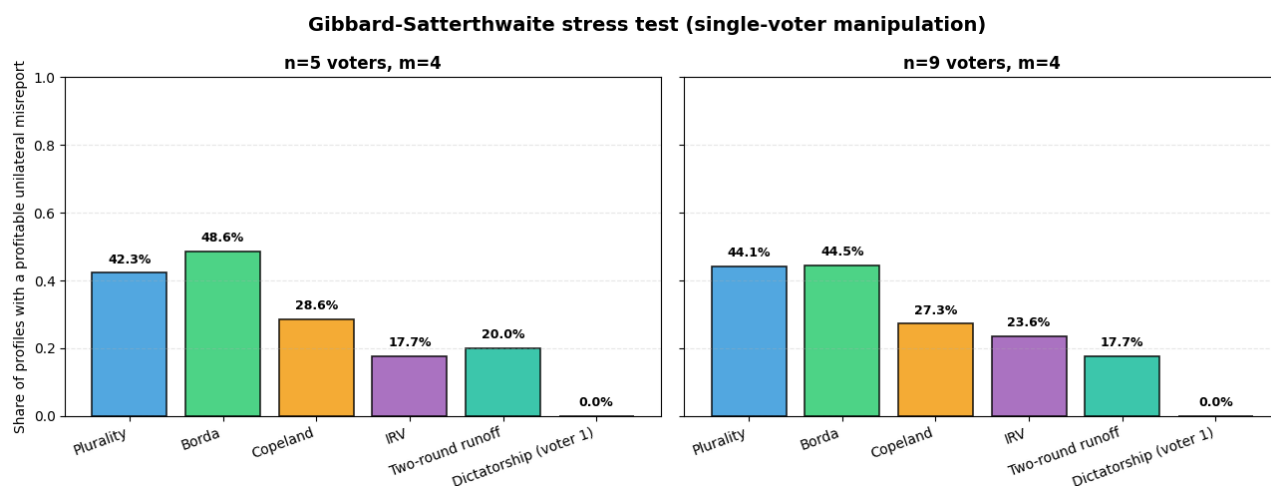


Figure 2: All non-dictatorial voting rules exhibit substantial rates of profitable single-voter manipulation, while dictatorship alone remains fully strategy-proof.

4 Beyond Arrow

Arrow shows that no aggregation rule can satisfy all of a set of “reasonable” conditions, those axioms we aspire to see reflected in our democratic mechanisms. Rather than regarding Arrow’s theorem, and other social choice theorems (Gibbard–Satterthwaite, May, etc.), nihilistically, modern democratic institutions structure themselves to effectively and transparently make the trade offs necessary when aggregating across an electorate’s individual preferences.

In their book *“Social Choice and Legitimacy: The Possibilities of Impossibility”*, coauthors John Patty and Elizabeth Penn argue that these theorems do not show that democratic decision-making is impossible or illegitimate, as some might posit. Instead, they argue that legitimacy cannot be grounded in any single, universal aggregation rule that satisfies all appealing axioms at once. There is no perfect or even a one-size-fits-all mechanism for voting.

Arrow demonstrates that no social welfare function can simultaneously satisfy a set of individually reasonable fairness conditions, while Gibbard–Satterthwaite shows that no non-dictatorial deterministic rule is immune to strategic manipulation. The authors take these results seriously but reject the conclusion that they undermine democracy. Instead, they argue that different institutional contexts prioritize different values (e.g., strategy-proofness, responsiveness, inclusiveness), and no mechanism can optimize all of them simultaneously. Under this framing, adopting election mechanisms becomes a series of tradeoffs to best match a preferred system to a society’s needs and values.

5 Conclusion

Instead of asking the question "Which voting rule is fair?", mechanism designers should ask "Under what conditions can outcomes produced by imperfect rules still be legitimate?". Legitimacy arises not from the fulfillment of our idealized axioms alone, but from procedural features such as participation, contestation, accountability, and transparency that permits citizens to recognize and accept tradeoffs. Impossibility theorems thus motivate a pluralist, context-dependent view of democratic design: impossibility results constrain what is achievable, but they also clarify the space of reasonable institutional choices.

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